



GE Medical Systems

Technical Publications

Direction 2401984
Revision 4

Signa® Saltbox Magnet Bridge Support.

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Service Documentation

TABLE OF CONTENTS

1.	Power Down & Lock Out Tag Out Pr ocedures	3
2.	Overview	6
3.	Modification Of Front End Bell For New Bri dge Support	7
4.	Modification Of Bridge For New Bridge Support	11
5.	Installation Of Bridge Support	12
6.	Revision Hi story	15

NOTE: Before shutting down the power, read through the entire procedure. Make sure the carriage is in the proper position for the bridge removal procedure.

NOTE: The following lock out tag out procedures should apply in most cases. However, if you have a system type not covered in this section, proceed to the safety section of the appropriate service documentation and follow the Lock Out Tag Out procedures.

1. Power Down & Lock Out Tag Out Procedures



POSSIBLE PERSONAL INJURY! AVOID SERIOUS INJURY OR DEATH BY ELECTROCUTION. REMOVE POWER FROM THE ACGD HARDWARE BEFORE WORKING ON THE MODULE. LOCK AND TAG OUT POWER TO THE ACGD HARDWARE SYSTEM WALL BOX DISCONNECT BEFORE STARTING WORK.

1. Bring the PC software down.

LX (SGI based) system software shutdown.

- a. Toggle mouse control to the PC monitor using the hard key on the top, left side of the keyboard housing.
- b. From the PC monitor use the mouse to select Start → Shutdown → Shut down.
Bring the Signa software down.
- c. Toggle mouse control back to the Signa monitor using the hardkey on the top, left side of the keyboard.
- d. Single-click on the Toolbelt icon.
- e. Single-click on the System Shutdown softkey on the Service Desktop Manager.
- f. Select OK to confirm the shutdown.

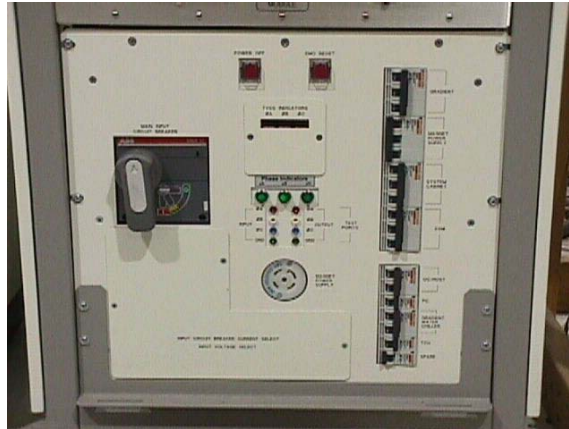
2. Wait for the system to indicate on the monitor that it is safe to power off the computer before proceeding. This usually takes about 90 seconds before this message is seen.

Horizon (Genesis console) system software shutdown.

Bring the Signa software down.

- a. Touch Utilities .

- b. Touch System Shutdown.
 - c. Select OK to confirm the shutdown.
 - d. Wait for Boot Prompt (this indicates it is safe to proceed).
3. Remove the cover from the front of the ACGD cabinet, See Illustration 1-1.



BREAKER LOCATION
ILLUSTRATION 1-1

4. Rotate the main breaker on the ACGD/PDU labeled INPUT counter-clockwise to the green, O (OFF) position. See Illustration 1-2.



MAIN INPUT BREAKER IN THE OFF POSITION
ILLUSTRATION 1-2

5. Verify that the 3 phase indicator lamps on the front of the power distribution unit (Illustration 1-5) are not illuminated

6. Lock Out / Tag Out the PDU "INPUT" circuit breaker.
 - a. Press on the left-pointing arrow on the end of the handle to expose the location where the lock and tag can be placed.
 - b. Lock and tag out the PDU "INPUT" circuit breaker as shown in Illustration 1-3



PDU MAIN INPUT CIRCUIT BREAKER LOCKED AND TAGGED OUT
ILLUSTRATION 1-3

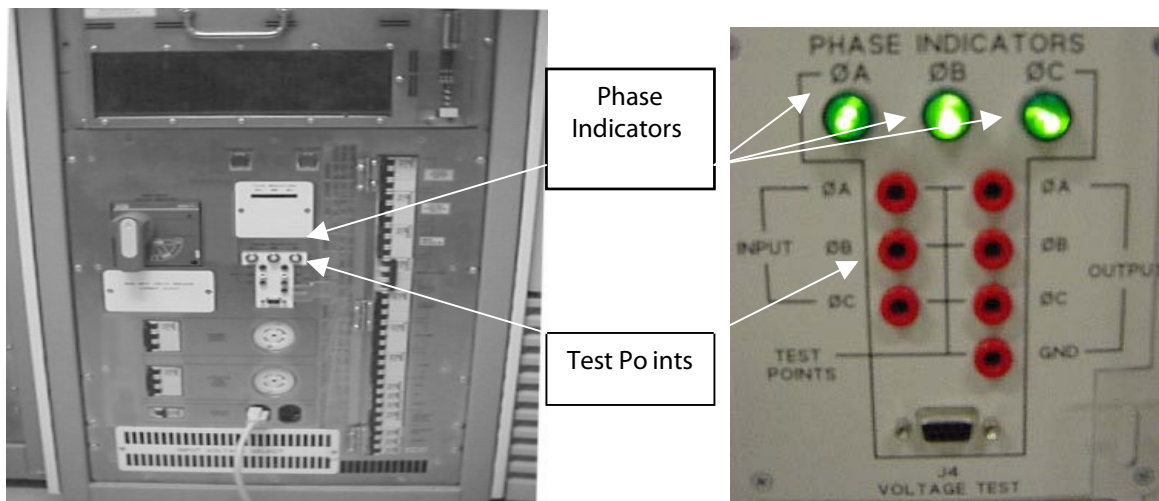


ILLUSTRATION 1-4

ILLUSTRATION 1-5
ACGD Cabinet –PDU (Front Cover Removed)

7. Verify that all energy has been dissipated. Measure the power at the front panel 420 and 208 test points of the PDUs. See Illustration 1-4. Voltages should now be at 0vac

Note : The test points are 100 :1 ratio (208 Vac = 2.08 Vac 420Vac = 4.20Vac)

2. Overview

Recent protocol developments are now forcing a tighter control of patient motion during scanning. The historical mechanical enclosure / bridge assembly is no longer an acceptable mechanical configuration to minimize patient motion. The M3090NE bridge support kit modifies the patient support to improve image quality. The bridge support kit contains a series of parts to modify any of the older style magnet /enclosure mechanical assemblies to match the current production product. This kit will simply prevent the forces generated by the gradient coil from moving the patient in an involuntary manner.

3. Modification Of Front End Bell For New Bridge Support

CAUTION

While the field engineer is drilling the holes, the second field engineer should turn on a vacuum cleaner and place the nozzle of the vacuum at the drilling point. This will help reduce the amount of dust in the air. Both field engineers must wear safety glasses and gloves during this step.

1. Prior to un-docking the table, move the LPCA to the rear of the magnet.
2. Un-dock the table and move to a safe location away from the front of the magnet.
3. Pull the front cover away from the magnet.
4. Remove enclosure cover to expose the existing bridge support.
5. Remove the two screws that secure the bridge to the front endbell. See Illustration 1. The screws will not be needed for the new configuration.

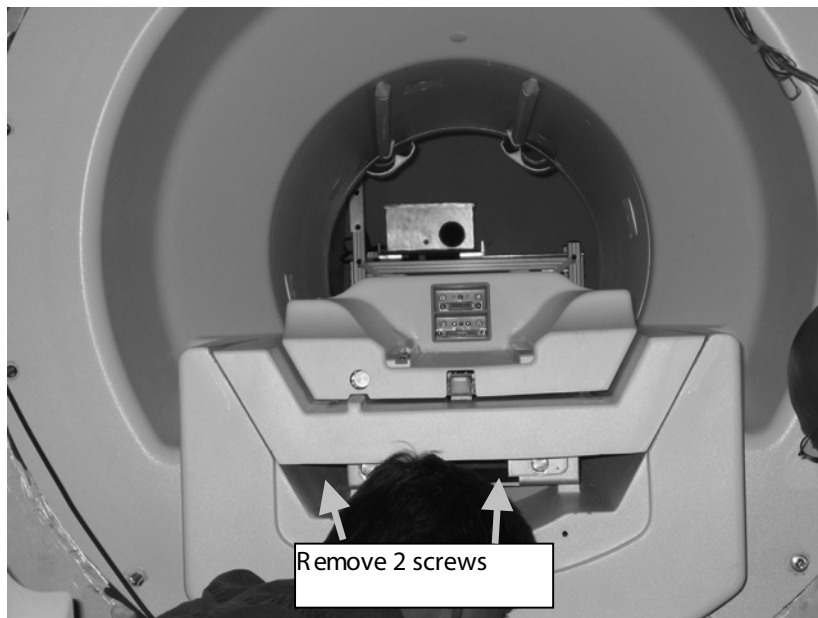


Illustration 1
Bridge Support removal

6. Remove the front endbell, and place it out of the magnet room.
7. Remove the bridge mounting bracket from the endbell. Place tape over the holes left in the endbell. See Illustration 2.

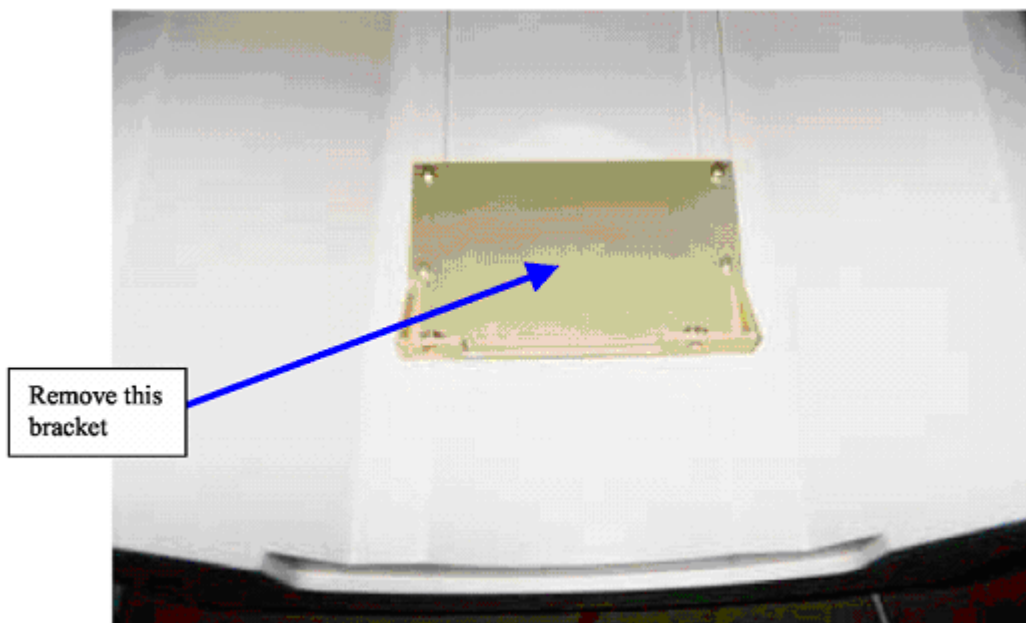


Illustration 2 Bracket Removal

8. Find the drill holes centerline on the front endbell.
 - a. Place the bridge support (shown in Illustration 10) against the front section of the endbell. See Illustration 3.
 - b. Scribe a line on the front endbell.
 - c. Measure $\frac{1}{2}$ inch or 13mm (half of the width of the bridge support) and scribe another line along the Z axis from the leading edge of the endbell.
 - d. This is where the holes need to be drilled.

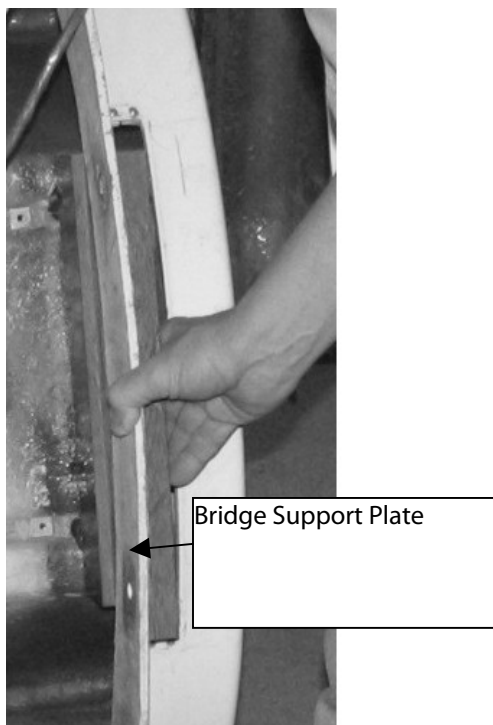


Illustration 3
Hole Alignment

9. Transfer the dimension achieved in step 7 from the bottom side of the endbell to the top of the endbell.
10. Find centerline on the bridge. Measure $5 \frac{1}{8}$ or 130 mm from the centerline and mark. See Illustration 4.

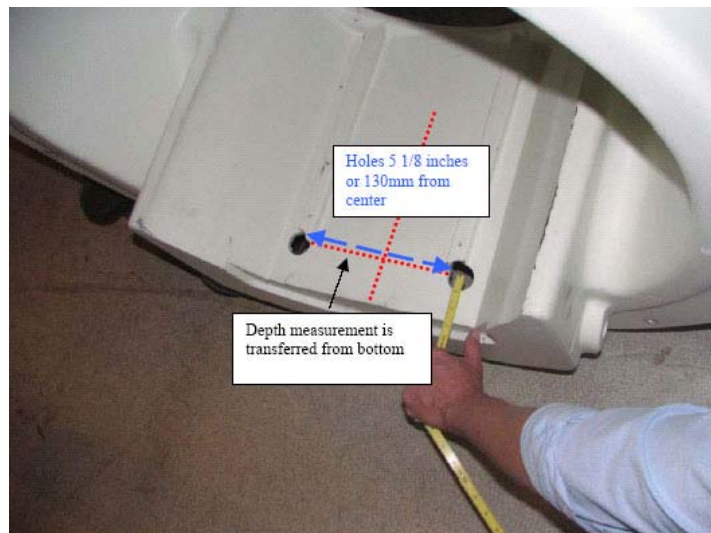


Illustration 4
Hole Location

10. Drill two 1.5 inch diameter holes centered at the locations marked in step 8. Use the drill bit (2353009) and with the hole saw (2353008).
11. Remove the mesh panel at the bottom of the front endbell. Using the cable saw (2353006) cut away the support in the middle of the air channel at the bottom of the endbell. See Illustration 5.



Illustration 5
REMOVAL OF AIR CHANNEL SUPPORT

4. Modification Of Bridge For New Bridge Support

1. Remove the cover from the LPCA.
2. Detach the drive belt from the LPCA and allow for slack in the belt in the front of the bridge.
3. Prop up the front of the bridge by placing any non-magnetic object between bridge and magnet. See Illustration 7.

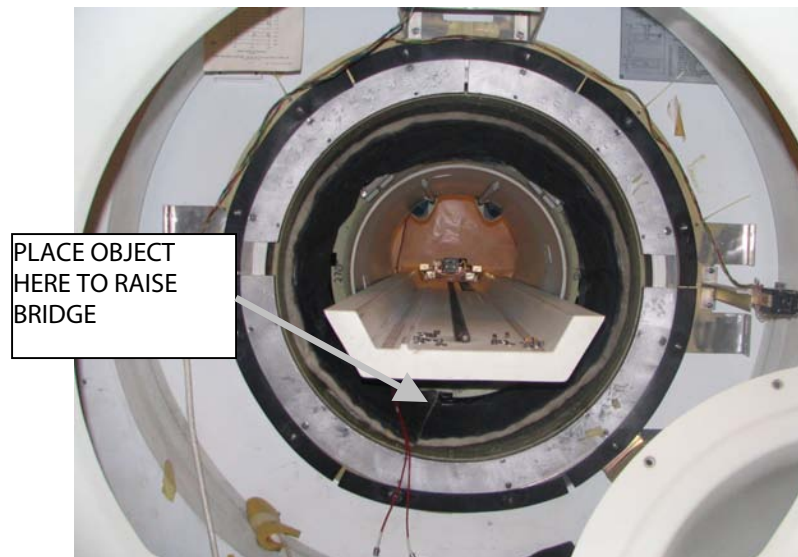


Illustration 7
Bridge Modification

4. Remove the original bridge mounting bracket from under the front part of the bridge. See Illustration 8. Note that the front pulley assembly is attached to the bridge mounting bracket.

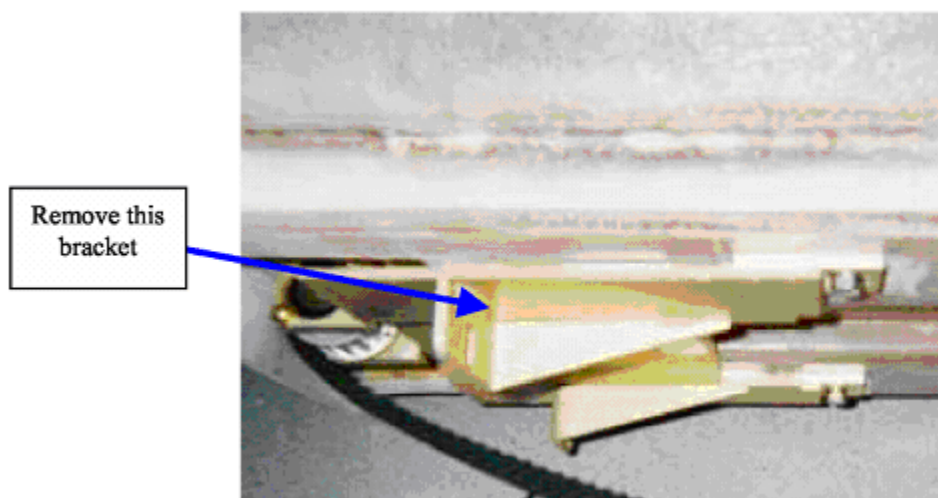


Illustration 8
Original Bracket on Bridge

5. Screw the front pulley wheel assembly to the stop plate (46-271431P1). See Illustration 9.

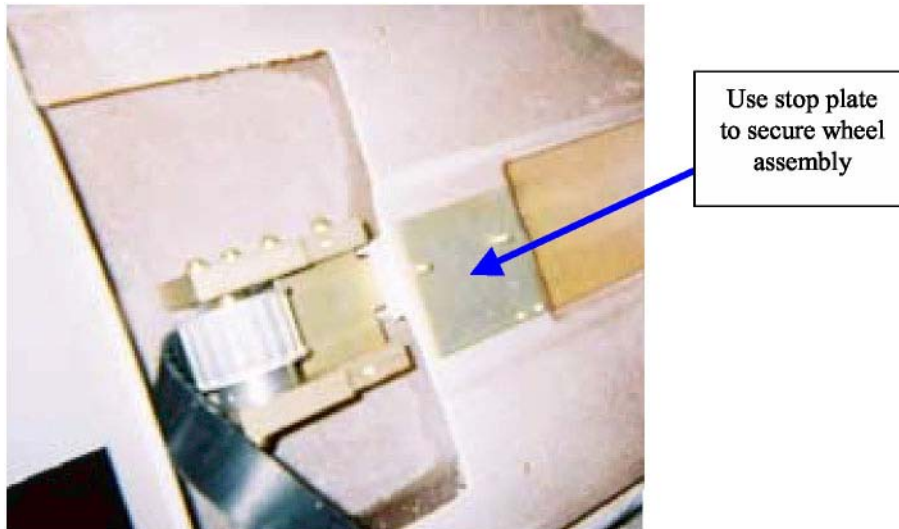


Illustration 9
Installation of new stop plate

5. Installation Of Bridge Support

1. Determine type of bridge support needed. See Illustration 10. The small racket is used for CX (either true CX or LCC with saltbox enclosure) magnets. The tall bracket is used for all S-series magnets.

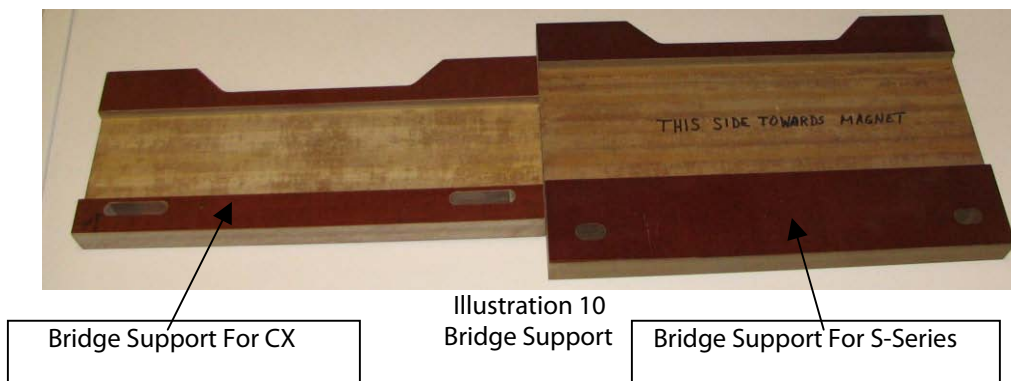


Illustration 10
Bridge Support

2. Install the threaded rod into the applicable bridge support as shown in Illustration 11 using the bridge support block, cup cap support 2318948, Inner bridge cap support 2345396 and stud, 2345395.



Illustration 11
Bridge Support Assembly

3. Re-install the front endbell, but do not install the 2 bottom bolts.
4. Position the bridge support in the proper location and install threaded rod into the two bottom holes as shown in Illustration 12.

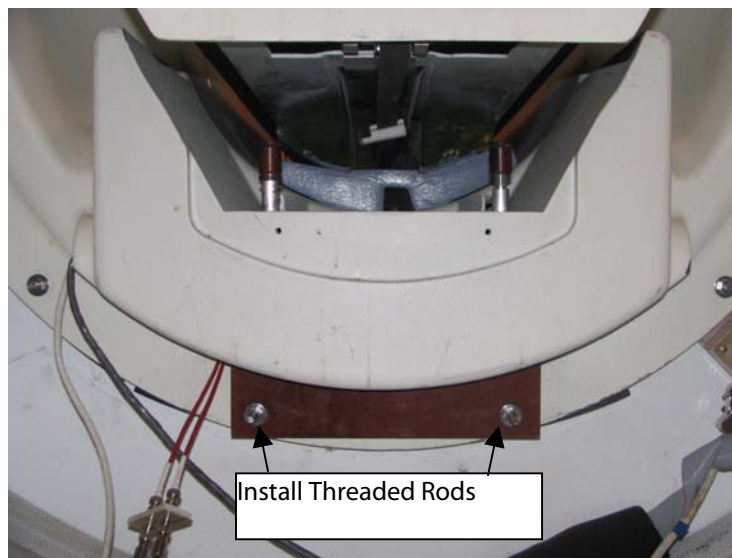


Illustration 12
Installation of Bridge Support

5. Adjust the bridge height so the bridge is approximately 1mm to 2mm above the front endbell on the right and left side. See Illustration 13.

Note: The final adjustment must ensure the bridge is not in physical contact with the body coil or endbell.

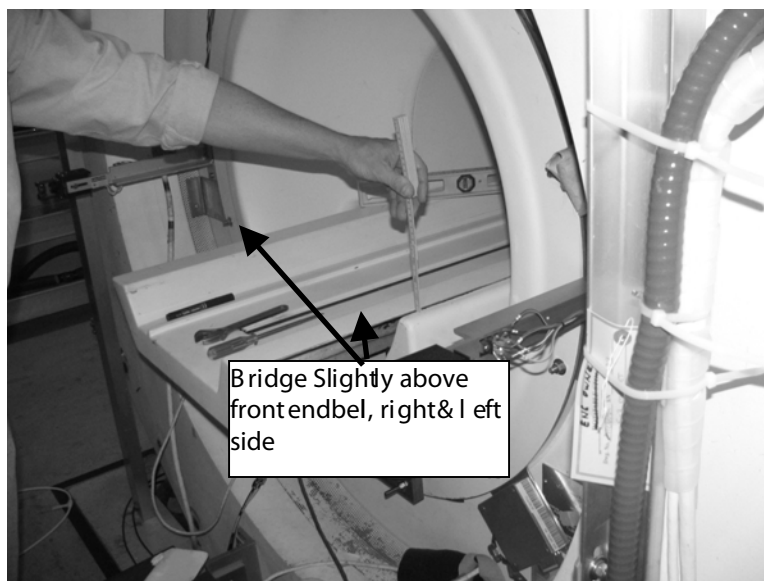


Illustration 13
Bridge Height Adjustment

6. Verify that the bridge is level front to back by checking with a long level. See Illustration 14.

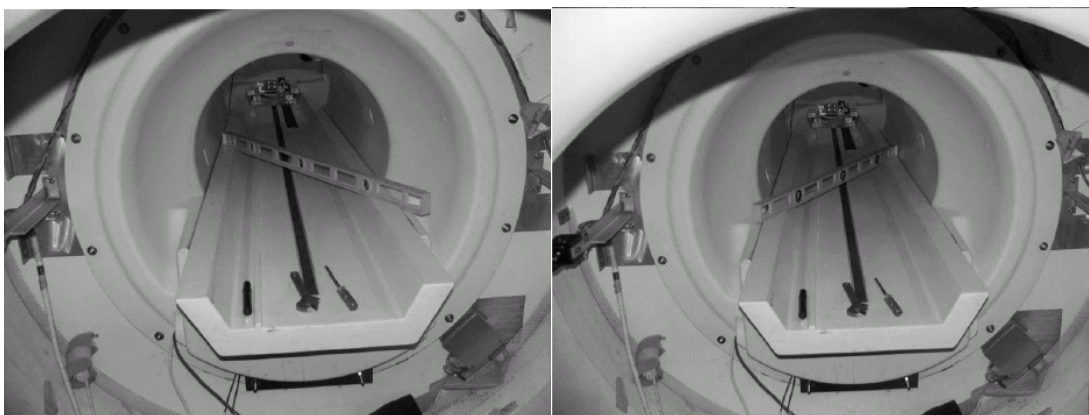


Illustration 14
Bridge Leveling

7. Return system to normal configuration.

7. Revision History

REV	DATE	AUTHOR	PRIMARY REASONS FOR CHANGE
0	April 29, 2004	P. Kargard	Initial release
1	Sept 16, 2004	P. Kargard	Update.
2	Mar 13, 2006	P. Kargard	Update
3	Jan.24, 2007	R. Stetz	Update: Note and Important message removed on Page 1. Now N/A
4	July18, 2007	M. Radziun	Update: Removed section 4 Modification of Front cover for new bridge.